

Improving the Computational Performance and Hydraulic Routing Fidelity of the Stormwater Management Model

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ABSTRACT

The Environmental Protection Agency's (EPA) Stormwater Management Model (SWMM) has been widely used by practitioners for various stormwater and collection system planning, operations, and design studies often driven by regulatory imperatives since its inception. Emerging applications including uncertainty characterization, multi-scenario evaluations, integrated optimization of operations and sizing and placement of water infrastructure systems including the design of in-line or deep tunnel systems, coastal flooding, emergency response, etc., increasingly require improved computational performance and fidelity of the hydraulic routing capabilities within SWMM. Of particular interest to practitioners is the movement towards smart or intelligent stormwater and collection systems, which envisions the fusion of SWMM with multimodal data from increasingly ubiquitous real time sensing technologies, and artificial intelligence and machine learning/data driven approaches to automatically and optimally manage collection systems. The contributions enumerated to support these applications in the paper are threefold and include: 1) Refactoring of the SWMM computational code to be "cache friendly", improving data locality in memory for improved computational performance; 2) a new virtual junction to explicitly handle momentum conservation to improve resolution of rapidly varied flow sub-pipe dynamics; and 3) and implementation of a recently proposed Dynamic Preissmann Slot (DPS) approach to improve stability

and handling of transitions between open-channel and pressurised flows. The resulting model is evaluated for computational performance, conservativeness, and stability through a comparison to SWMM and to various analytical solutions. Finally, the new formulations evaluated on a model representing collection system for the City to evaluate flooding and its implication for pollutant transport and fate.

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- a journal article in press (Dasgupta 2008),
- an article in an edited book using @INCOLLECTION (Zadeh 1981),
- a building code using @MANUAL (International 1988),
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- a doctoral thesis using @PHDTHESIS (Chang 1987),
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- a paper in a proceedings using @INPROCEEDINGS (Eshenaur et al. 1991; Garrett 2003),
- a standard using @INCOLLECTION (ASTM 1991),
- a translated book (Melan 1913),
- a two-part paper (Frater and Packer 1992a; Frater and Packer 1992b),
- a university report using @TECHREPORT (Duan et al. 1990),
- an untitled item in the Federal Register using @MANUAL (Federal 1988),
- works in a foreign language (Duvant and Lions 1972; Reiffenstuhl 1982),
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```
@MISC{Burka:1993a,  
    author = {Burka, L. P.},  
    title = {A hypertext history of multi-user dimensions},  
    journal = {MUD history},  
    year = {1993},  
    month = {Dec. 5, 1994},  
    url = {http://www.ccs.neu.edu}  
}
```

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```
howpublished = {CD-ROM},
```

APPENDIX I. NOTATION

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D = pile diameter (m);

R = distance (m); and

$C_{\text{Oh no!}}$ = fudge factor.

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- 334 7. The option `SectionNumbers` produces an automatic numbering of sections. Without the
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336 formatting in ASCE journals (note that the Appendixes will, however, be automatically
337 “numbered” with Roman numerals). With the `SectionNumbers` option, sections and
338 subsections are numbered with Arabic numerals (e.g. 2, 2.1, etc.), but subsubsection
339 headings will not be numbered.
- 340 8. The options `NoLineNumbers`|`LineNumbers` can be used to override the default use (or
341 absence) of line numbers in the `Journal`, `Proceedings`, and `NewProceedings` formats.

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408

List of Tables

409

1

An example table

21

TABLE 1. An example table

Assembly Attribute (1)	Values (2)
Number of particles	4008
Particle sizes	Multiple
Particle size range	$0.45D_{50}^*$ to $1.40D_{50}$
Initial void ratio, e_{init}	0.179
Assembly size	$54D_{50} \times 54D_{50} \times 54D_{50}$
* D_{50} represents the median particle diameter	

410	List of Figures	
411	1	An example figure (just a box). This particular figure has a caption with more
412		information than the figure itself, a very poor practice indeed. A reference here
413		(Stahl et al. 2004). 23

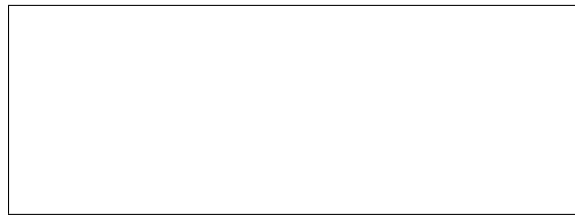


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